

A Technical Introduction to Autoregressive model

Section 1

It is seen that X_t is white noise convolved with the ϕ_k kernel plus the constant mean. If the white noise ε_t is a Gaussian process then X_t is also a Gaussian process. In other cases, the central limit theorem indicates that X_t will be approximately normally distributed when ϕ is close to one. For $\varepsilon_t = 0$, the process $X_t = \phi X_{t-1}$ will be a geometric progression (exponential growth or decay). In this case, the solution can be found analytically: $X_t = a \phi^t$ whereby a a is an unknown constant (initial condition).

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$\gamma_m = \sum_{k=1}^m \phi_k \gamma_{m-k} + \sigma_{\varepsilon}^2 \delta_{m,0}$, $\delta_{m,0} = 1$ if $m=0$ and 0 otherwise.

Section 3

which can be solved for all ϕ_m ; $m = 1, 2, \dots, p$. The remaining equation for $m = 0$ is

Section 4

An alternative formulation is in terms of the autocorrelation function. The AR parameters are determined by the first $p+1$ elements $\rho(\tau)$ of the autocorrelation function. The full autocorrelation function can then be derived by recursively calculating

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$X_{t+1} = X_t + (1 - \theta)(\mu - X_t) + \varepsilon_{t+1}$, where $|\theta| < 1$, $\mu := E(X)$ is the model mean, and $\{\varepsilon_t\}$ is a white-noise process with zero mean and constant variance σ^2 . By rewriting this as $X_{t+1} = \theta X_t + (1 - \theta)\mu + \varepsilon_{t+1}$ and then deriving (by induction) $X_{t+n} = \theta^n X_t + (1 - \theta^n)\mu + \sum_{i=1}^n (\theta^{n-i} \varepsilon_{t+i})$, one can show that

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where B is the backshift operator, where $\phi(\cdot)$ is the function defining the

autoregression, and where ϕ_k are the coefficients in the autoregression. The formula is valid only if all the roots have multiplicity 1. The autocorrelation function of an AR(p) process is a sum of decaying exponentials.

Section 7

Estimation of autocovariances or autocorrelations. Here each of these terms is estimated separately, using conventional estimates. There are different ways of doing this and the choice between these affects the properties of the estimation scheme. For example, negative estimates of the variance can be produced by some choices. Formulation as a least squares regression problem in which an ordinary least squares prediction problem is constructed, basing prediction of values of X_t on the p previous values of the same series. This can be thought of as a forward-prediction scheme. The normal equations for this problem can be seen to correspond to an approximation of the matrix form of the Yule–Walker equations in which each appearance of an autocovariance of the same lag is replaced by a slightly different estimate. Formulation as an extended form of ordinary least squares prediction problem. Here two sets of prediction equations are combined into a single estimation scheme and a single set of normal equations. One set is the set of forward-prediction equations and the other is a corresponding set of backward prediction equations, relating to the backward representation of the AR model:

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When the polynomial division on the right side is carried out, the polynomial in the backshift operator applied to ε_t has an infinite order—that is, an infinite number of lagged values of ε_t appear on the right side of the equation.

Section 9

Example: An AR(1) process An AR(1) process is given by: $X_t = \phi X_{t-1} + \varepsilon_t$, where $\{\varepsilon_t\}$ is a white noise process with zero mean and constant variance σ_{ε}^2 . (Note: The subscript on ϕ_1 has been dropped.) The process is weak-sense stationary if $|\phi| < 1$ since it is obtained as the output of a stable filter whose input is white noise. (If $\phi = 1$ then the variance of X_t depends on time lag t , so that the variance of the series diverges to infinity as t goes to infinity, and is therefore not weak-sense stationary.) Assuming $|\phi| < 1$, the mean $E(X_t)$ is identical for all values of t by definition of weak sense stationarity. If the mean is denoted by μ

$\{\mu\}$, it follows from $E(X_t) = \phi E(X_{t-1}) + E(\varepsilon_t)$, that $\mu = \phi \mu + 0$, and hence

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The terms involving square roots are all real in the case of complex poles since they exist only when $\phi^2 < 0$. Otherwise the process has real roots, and:

Section 11

where $\phi_1, \dots, \phi_p$ are the parameters of the model, and ε_t is white noise. This can be equivalently written using the backshift operator B as

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which, once $\{\phi_m; m = 1, 2, \dots, p\}$ are known, can be solved for σ_{ε^2} .